Ex. N0:2

8 Queens Problem

AIM:

To write a program to solve 8 queens problem using SWI Prolog.

ALGORITHM:

Step 1: Start the program.

Step 2: Call solve(Board) to find a solution for 8 queens.

Step 3: Initialize the number of queens to 8.

Step 4: If the number of queens is 0, return an empty list.

Step 5: Otherwise, perform the following:

1. Decrease the number of queens by 1.
2. Recursively generate the positions of the remaining queens.
3. Select a row number between 1 and 8 for the current queen.
4. Check whether the current queen is safe from the other queens.

Step 6: Check the safety conditions:

1. Ensure that no two queens are in the same row.
2. Calculate the absolute difference between their row positions.
3. Ensure that the row difference is not equal to the column distance, preventing diagonal attacks.

Step 7: If the safety conditions are satisfied, continue checking the remaining queens.

Step 8: If any condition fails, use backtracking to try another row position.

Step 9: When all 8 queens are placed safely, display the solution.

Step 10: If another solution is requested, backtrack and search for the next valid arrangement.

Step 11: Stop when no more solutions are available.

PROGRAM:

```
solve(Board) :-
   queens(8, Board).

queens(0, []).

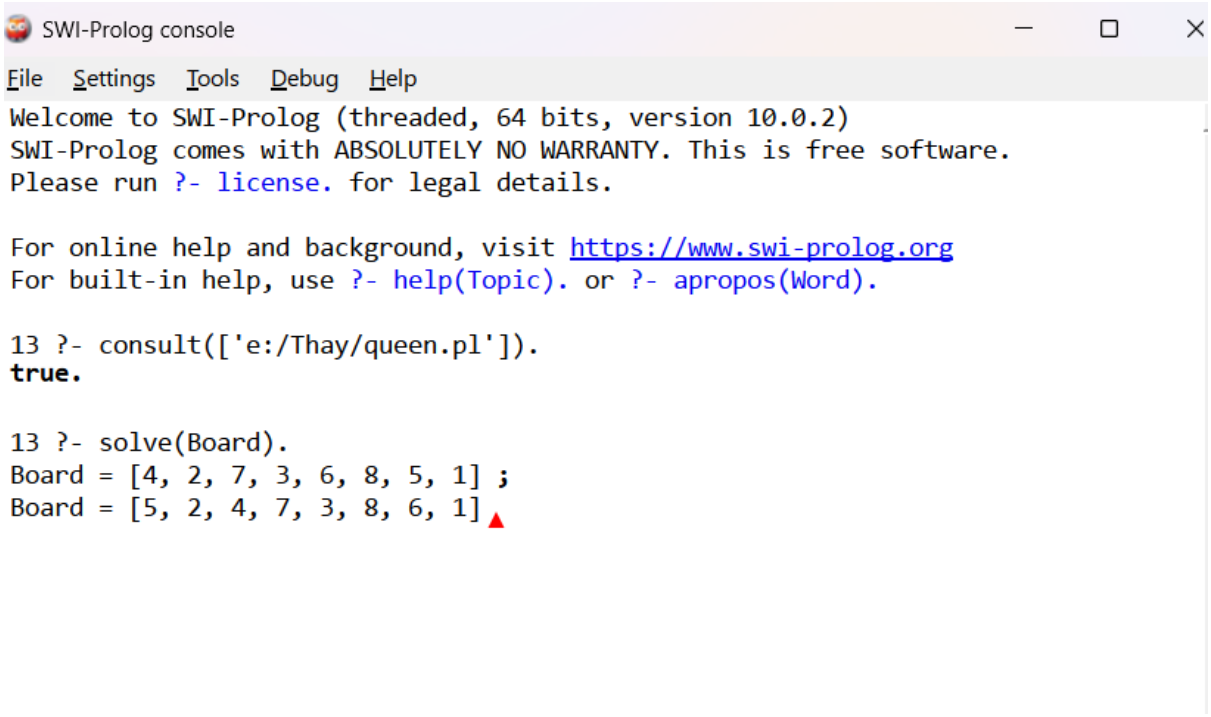
queens(N, [X|Rest]) :-
   N > 0,
   N1 is N - 1,
```

```
    queens(N1, Rest),
    between(1, 8, X),
    safe(X, Rest, 1).

safe(_, [], _).

safe(X, [Y|Rest], D) :-
    X =\= Y,
    A is abs(X - Y),
    A =\= D,
    D1 is D + 1,
    safe(X, Rest, D1).
```

OUTPUT:



RESULT:

Thus the 8 queens problem was executed successfully in SWI Prolog console and verified.